Long-lived Particles: Theoretical Motivations and Experimental Detections

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Abstract

The Standard Model (SM) of particle physics, despite its success, leaves open questions such as the nature of dark matter and the origin of neutrino masses. Many theories Beyond the Standard Model (BSM) predict the existence of Long-Lived Particles (LLPs) with lifetimes sufficient to travel macroscopic distances before decaying. This report provides a comprehensive review of the theoretical motivations and experimental strategies for detecting LLPs. I first summarize the physical mechanisms that suppress decay widths, including small couplings, heavy mediators, and phase space suppression. I then survey various BSM frameworks that naturally yield LLPs, such as Supersymmetry, Higgs-portal models, and Heavy Neutrino theories, and introduce the "simplified model" approach used to categorize their production and decay modes. Finally, we discuss the current experimental techniques at the LHC, detailing direct search methods utilizing Time-of-Flight (ToF) and ionization energy loss (dE/dx), as well as indirect searches focusing on displaced objects and missing transverse energy.

1 Introduction

The Standard Model of particle physics has been remarkably successful in describing the fundamental particles and their interactions. Though we have found Higgs Boson in 2012 which filled in the final piece of the particle spectrum in the SM, it leaves several questions unanswered, such as the nature of dark matter, the baryon asymmetry of the universe, the hierarchy problem, and the origin of neutrino masses.

Lifetimes of the particles in the SM span a wide range, from the Z boson with a lifetime of approximately 2×10^{-25} seconds to the proton, which is predicted to have a lifetime exceeding 10^{34} years.

Models beyond the Standard Model (BSM) also predict new particles with a variety of lifetimes. Some of these particles may be long-lived, meaning they have lifetimes long enough to travel macroscopic distances before decaying. For particles approaching the speed of light, this can lead to detectable displacements between their production and decay points for $c\tau \gtrsim 10\mu m$[1].

In the past decades, the primary search at the Large Hadron Collider (LHC) has focused on promptly decaying particles which decay almost immediately after production at the interaction point, producing high transverse momentum (p_T) leptons, photons and jets. However, we have not observed any significant deviations from the SM predictions in these searches. That motivates us to move from strongly-coupled prompt paticles to alternative scenarios, such as long-lived particles (LLPs). The decay length of LLPs can range from micrometers to kilometers, even reaching the bounds set by Big Bang Nucleosynthesis (BBN)[11]. And what counts as an LLP depends on the context of a particular experiment. A short-lived particle in one experiment might be considered as an LLP to another, cause it depends on the experiment's size and energy scale[21].

In this report, we will review the theoretical motivations for LLPs and experimental search of new LLPs.

2 Particles Lifetime

2.1 Definition of Lifetime

Fundational particles have a fixed decay probability at every single moment, which leads to a Poisson distribution to describe the probability of a given particle decaying within a fixed time interval. In turn, the probability of a particle surviving after a time t, measured in the lab frame, is given by an exponential distribution:

$$P(t) = Ce^{-t/\gamma\tau} \tag{1}$$

where τ is the lifetime of the particle, as well as the mean of the exponential distribution, and $\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$ is the Lorentz boost of the particle. So the lifetime τ represents the time when a particle has a $1/e$ chance of not having decayed in its own rest frame.

2.2 Decay Width and Lifetime

Lifetime is a fundamental property of a particle, but it differs from some other properties like charge and spin. The latter are intrinsic and unchanging, which define the particle. However, lifetime can depend on the particle's masses, charges, spins and the couplings with other particles. The decay width Γ of a particle is inversely related to its lifetime τ by the relation:

$$\tau = \frac{\hbar}{\Gamma} \tag{2}$$

A larger decay width corresponds to a shorter lifetime, indicating that the particle decays more rapidly. And the decay width is determined by particles and their interactions and can be given by:

$$\Gamma = \frac{1}{2m} \int d\Pi_f |\mathcal{M}(i \to f)|^2 (2\pi)^4 \delta^4(p_i - p_f) \tag{3}$$

where m is the mass of the decaying particle, $d\Pi_f$ is the phase space element of the final states, $\mathcal{M}(i \to f)$ is the matrix element for the decay process, and the delta function ensures conservation of four-momentum.

2.3 What are LLPs and Why they have long lifetime

If in a given experiment, a particle's lifetime is long enough that it can travel a measurable distance larger than the experiment resolution before decaying, it is classified as an LLP. This depends on the time and position resolution of the detector as well as the relativistic boost of the particle. At the LHC, particles with lifetimes greater than about 1 ps are generally considered long-lived[21].

According to Eq.3, there are several ways a particle can achieve a long lifetime:

- **Small Couplings:** If the particle interacts very weakly with other particles, the matrix element $\mathcal{M}(i \to f)$ will be small, leading to a small decay width and thus a long lifetime.
- **Heavy Mediators:** If the interaction has a heavy mediators with $M \gg m$, the heavy scale will show up in the denominator of the propagator in calculating $\mathcal{M}(i \to f)$ and suppresses the matrix element, then the effective coupling can be suppressed.
- **Phase Space Suppression:** If the integral over $d\Pi_f$ is small, the decay width will be small, which corresponds to the situation where the mass of the decaying particle is only slightly larger than the total mass of its decay products, the available phase space for the decay will be limited.

Also there are other reasons why LLPs have long lifetime. For instance, certain **symmetries** can forbid or suppress specific decay channels. An example is the proton, which is stable in the Standard Model due to baryon number conservation. And a fundamental particle's lifetime can differ in a bound state.

3 Theories that predict new LLPs

Long-lived particles exists in many well-motivated BSM theories, such as the minimal supersymmetric SM (MSSM), and newer frameworks like neutral naturalness[8] and hidden sector DM[7]. Now we will briefly review some categories of the theories that predict new LLPs.(you can refer to [11] for more details) Actually, for a given model, it may belong to multiple categories below. They should not be considered as exclusive categories.

3.1 Supersymmetry-like theories

This category contains models with multiple new particles carrying SM gauge charges and a variety of allowed cascade decays.

- **approximate symmetries:** R-parity[5] or SUSY itself in the case of gauge mediation[13](because the next-to-lightest supersymmetric particles (NLSP) decay coupling to the gravitino is inversely proportional to the high supersymmetry breaking scale).
- **hierarchy of mass scales:** highly off-shell intermediaries in split SUSY[3] causes heavy mediators, or nearly-degenerate multiplets[9] in anomaly-mediated SUSY breaking[15] causes phase space suppression.
- **SUSY hidden sectors:** stealth SUSY[14].

This categoriy includes any non-SUSY models with new SM gauge-charged particles, because they all predict new particles above the weak scale with SM gauge charges. In this category, LLP production is typically dominated by SM gauge interactions.

3.2 Higgs-portal theories

In Higgs-portal theories, LLPs couple mainly to the SM-like Higgs boson, because the SM Higgs field provides one of the leading renormalizable portals for new gauge-singlet particles to couple to the SM, and the experimental of the Higgs boson leaves much available space for couplings of BSM particles to the Higgs[10]. The coupling may be suppressed by small mixing angles or small Yukawa couplings, leading to long lifetimes.

For example, in many Hidden Valley (HV) scenarios[26], there are exotic Higgs decays to low-mass particles[12]. This can happen in many well-motivatied models, such as models of neutral naturalness, DM and relaxion models[6].

In many scenarios where LLPs are produced in exotic Higgs decays, associated-production modes can be the only way of triggering on the event[1]. And except for gluon-fusion process, the Higgs often comes with many associated production modes like vector-boson fusion (VBF) and Higgs-strahlung (VH) production modes

3.3 Gauge-portal theories (ZP)

This category contains scenarios where new vector mediators can produce LLPs. These are similar to Higgs models. There are models like:

- both SM fermions and LLPs carry a charge associated with a new Z'[23].
- either Abelian or non-Abelian dark photon or dark Z models[20] with the couplings of new vector bosons to the SM are mediated by kinetic mixing.

So the long lifetime comes from small kinetic mixing or phase space suppression.

These are well motivated by theories of DM, particularly models with significant self-interactionsbor sub-weak mass scales[1].

3.4 Dark Matter (DM) theories

This category collects Non-SUSY and hidden-sector DM scenarios. An explicit detector-level signature of a DM candidate, i.e. missing energy, is an important feature to distinguish this category from Higgs-portal theories and Gauge-portal theories above.

Examples of these include models:

- new electroweak multiplets[27].
- strongly interacting massive particles (SIMPs)[19].
- inelastic DM (iDM)[25]. Generically, iDM produce two dark states that are very close in mass, $\Delta M = M_{\chi_2} - M_{\chi_1} \ll M_{\chi_1}$. So the phase space of decay $\chi_2 \longrightarrow \chi_1 + q\bar{q}$ will be suppressed.
- models with DM coannihilation partners[22]. Lead to long lifetimes through phase space suppression.
- non-thermal "freeze-in" scenarios[17].

3.5 Heavy neutrino theories (RHν)

The seesaw mechanism of SM neutrino mass generation predicts new right-handed neutrino (RHN) states[17]. If the RHNs have masses in the GeV to TeV range, they can be probed at the LHC that they are LLPs[18]. Examples of well-motivated, UV-complete models with RHNs include:

- the neutrino minimal SM (MSM)[4].
- the left−right symmetric model[24].

For example, in the low-scale seesaw neutrinos model, there exist two or more RHNs that couple to the left-handed neutrinos (LHNs). Because of the seesaw mechanism, the SM neutrinos are light when the sterile neutrinos are extremely heavy, or sterile neutrinos have smaller couplings. For a sterile neutrino N mainly consisted of a RHN, it can feel the weak interaction but smaller than the SM neutrinos. So if a SM process produces the SM neutrino, such as $W^+ \longrightarrow \ell^+\nu$, there is a small probability that $W^+ \longrightarrow \ell^+ N$ happens instead. In turn, if the N decay back $N \longrightarrow W^+\ell^-$ and its mass is smaller than the W boson, the heavy mediator will suppress the decay width of N and lead to a long lifetime.

3.6 Simplified models yielding LLPs

Given the large number of theories predicting LLPs, it is useful to introduce some simlified models[1]. The simplified models are organized in terms of LLP channels characterized by a combination of a particular LLP production mode with a particular decay mode. Because of the long lifetime, production and decay positions of LLPs are physically distinct, which means they can often be factorized, allowing separate consideration of production and decay modes. For each LLP channel, the lifetime of the LLP is taken to be a free parameter.

Actually, simlified models have their own limits[2]. For example, the presentation of search results in terms of simplified models often assume 100% branching fractions into particular final states. Thus, in a UV model where the LLP decays in a very large number of ways, none of the individual simplified model searches may be sufficient to constrain it.

3.6.1 Production Mode

There is minimal set of production modes[1] for LLPs and schematic diagrams for each production mode are shown in figure 1.

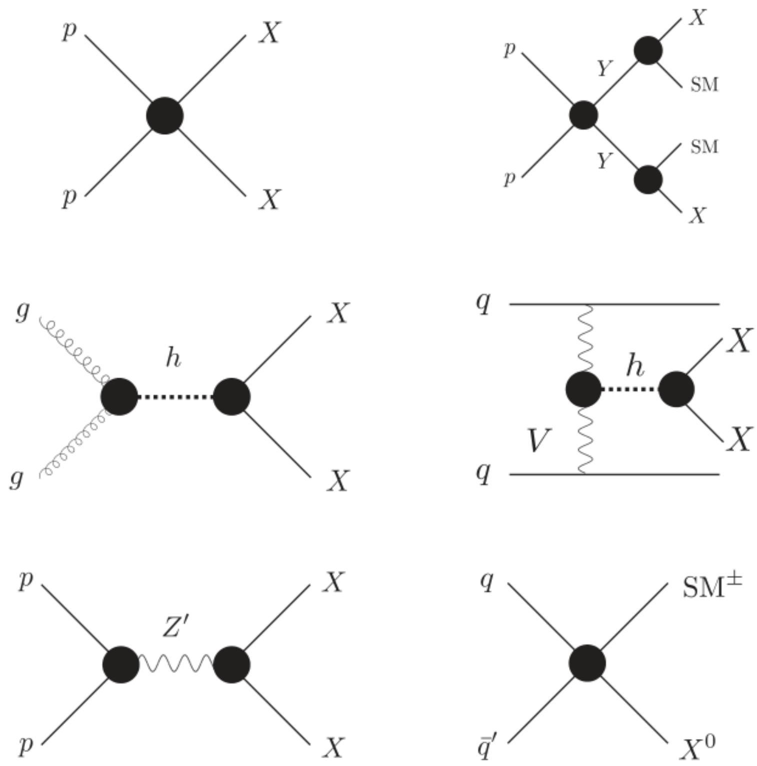

Figure 1: Schematic illustrations of LLP production modes in our simplified model framework. From top to bottom and left to right: direct pair production (DPP); Heavy parent (HP); Higgs modes (HIG), including gluon fusion and VBF production (not shown here is VH production); heavy resonance (RES); charged current (CC).

These production modes determine LLP signal rates by determining the LLP production cross section and kinematic distributions, and make clear prediation for the number and type of prompt objects accompanying the LLPs. Below we briefly describe each production mode. Some models may span several production modes.

- **Direct-pair production (DPP):** the LLP is dominantly pair-produced non-resonantly from SM initial states.
- **Heavy parent (HP):** the LLP is produced in the decays of on-shell HP particles that are themselves pair produced from the pp initial state.
- **Higgs (HIG):** the LLP is produced through its couplings to the SM-like Higgs boson. This includes gluon-fusion, VBF, and VH production modes.
- **Heavy resonance (RES):** the LLP is produced in the decay of an on-shell resonance, such as a heavy Z' gauge boson initiated by $q\bar{q}$ initial state.
- **Charged current (CC):** In models with weak-scale RHNs, the LLP can be produced in the leptonic decays of W/W'.

3.6.2 Decay Mode

There is an irreducible set of decay modes that can be used to characterize LLP signatures for various LLP charges (including neutral, electrically charged, and color charged)[1].

- **Di-photon decays ($\gamma\gamma + inv.$):** the LLP can decay resonantly to $\gamma\gamma$ (like in Higgs-portal models or left−right symmetric models) or to $\gamma\gamma$ + invisible (in DM models).
- **Single-photon decays ($\gamma + inv.$):** the LLP decays to γ + invisible (like in SUSY models).
- **Hadronic decays ($jj(+inv.)$):** the LLP can decay into two jets (jj) (like in Higgs and gauge-portal models, or RPV SUSY), jj + invisible (SUSY, DM, or neutrino models), or j + invisible (SUSY). Here, "jet" (j) means either a light-quark parton, gluon, or b-quark.
- **Semi-leptonic decays ($jj\ell$):** the LLP can decay into a lepton + 1 jet (such as in leptoquark (LQs) models) or 2 jets (like in SUSY or neutrino models).
- **Leptonic decays ($\ell^+\ell^-(+inv.)$):** the LLP can decay into $\ell^+\ell^-$ + invisible , or $\ell^\pm$ + invisible (as in Higgs-portal, gauge-portal, SUSY, or neutrino models). Here the symbol ℓ may be any flavor of charged lepton, but the decays are lepton flavor-universal and (for $\ell^+\ell^-$ decays) flavor-conserving.
- **Flavored leptonic decays ($\ell^+_\alpha\ell^-_{\beta\neq\alpha}(+inv.)$):** the LLP can decay into ℓ_α + invisible, $\ell^+_\alpha\ell^-_{\beta\neq\alpha}$, or $\ell^+_\alpha\ell^-_{\beta\neq\alpha}$ + invisible (as in SUSY or neutrino models).

In all cases, both the LLP mass and proper lifetime are free parameters.

3.6.3 Specific Smiplified Models

The smiplified models can be classified into three categories according to the LLPs' gauge charges:

- **Neutral LLPs**.
- **Electrically charged LLPs** $|Q| = 1$.
- **LLPs with color charge**.

More details about the specific simplified models can be found in [1].

4 Experimental Search of new LLPs

4.1 Direct searches for new LLPs

Direct searches for new LLPs today use modern technologies to look for particles interacting in the detector in ways that are inconsistent with any SM particle. At the LHC, $\gamma\tau \geq \mathcal{O}(0.1)$ ns is a convenient threshold beyond which direct detection can be reasonably efficient[21].

For new LLPs that are electrically (or occasionally magnetically) charged, we can use direct searches to distinguish the track of a new LLP from an SM LLP. We usually focus on tracks that are sensitive to mass or charge, kinematics, or decay position. Also, for LLPs charged under strong force, hadronic interactions with detector material could make direct detection possible

When charged particles undergo helical motion in a magnetic field, tracks are usually fit with a 5-dimensional helix. A common parametrization of the helix includes charge-to-momentum ratio of a track ($\frac{q}{p}$), two angles of the track ($\phi\&\theta$), and the longitudinal and transverse impact parameters ($z_0\&d_0$). With these 5 parameters, we can get the vast majority of collider measurements and searches. However, to determine the mass of a particle from its track, we need more information, such as β, which can be measured by time-of-flight or energy loss per unit length ($\frac{dE}{dx}$).

We can measure the Time-of-Flight (ToF) to get β, which is the time it takes for a particle to reach a given detector layer, assuming that the particle was produced at the origin of the detector at $t = 0$. Comparing the ToF for a $v = c$ particle to the same detector position to our measured ToF allows us to calculate β. To do this, the time resolution of the detector must be significantly better than the time differences we want to measure. A detector at a distance d from the origin must have timing resolution σ_t satisfying:

$$\Delta t = \frac{d}{c}(\frac{1}{\beta} - 1) \geq n \cdot \sigma_t \tag{4}$$

Many detectors can also measure the energy loss per unit length ($\frac{dE}{dx}$) of charged particles as they pass through the detector material. And it can be given by the Bethe-Bloch formula[16]:

$$-\frac{dE}{dx} = 0.3071q^2 \frac{Z}{A} \frac{1}{\beta^2} \left[\ln \left(\frac{2m_e c^2 \beta^2 \gamma^2}{I} \right) - \beta^2 + \frac{\delta}{2} \right] \frac{\text{MeV}}{\text{g/cm}^2} \tag{5}$$

where q is the incident particle charge in units of the electron charge, Z and A are the atomic number and weight of the detector material, me is the electron mass, I is the mean excitation energy of the material, and δ is a density parameter that reduces the ionization for very relativistic particles. For a given material and particle charge, $\frac{dE}{dx}$ depends mainly on β. So measuring $\frac{dE}{dx}$ gives us another way to determine β.

4.2 Indirect searches for new LLPs

If an LLP is electrically neutral and has no strong charge, direct detection becomes very challenging cause it will not interact with the detector as it travels. And if it is stable and long-lived enough to decay outside the detector, the only signature of the LLP is missing momentum or missing transverse energy (MET).

For the LLPs that decay within the detector, we can search for their decay products to do indirect detection. At the LHC, $\mathcal{O}(0.1)$ ps $\leq \tau \leq \mathcal{O}(30)$ ns is a rough lifetime range in which indirect detection is possible[21]. And indirect detection relies on identifying one or more of the decay products as displaced from the primary collision in either space or time (or both).

If an LLP decay includes one visible particle and one neutral, weakly or non-interacting particle, we can search for a displaced object. If the LLPs are pair-produced and both decay in the detector, we can search for two displaced objects.

If the LLP decay includes a photon happens before the calorimeter, we can utilize the spatial resolution of the electromagnetic calorimeter to tag whether the photon point back to the interaction point, and utilize the timing resolution of the calorimeter to tag whether the photon arrives later than a photon produced in the primary interaction. These can tell whether the photon can be identified as " non-prompt" . The electron can also be identified as non-prompt in a similar way. If the LLP decay inside the calorimeter, the energy deposits from an LLP can produce unconventional shower shapes relative to prompt jets, cause this jet jet starts at some point inside the calorimeter rather than at the collision point.

For LLP decays that include a charged particle, this particle can produce a track. If the track originates after the first layer, we can well reconstruct a track with no hits in the first few layers, the we can identify this track as displaced. Also if the track has different impact parameters with the primary vertex of the main collision point in the event, we can identify this track as displaced. Thirdly, if the primary LLP is moving slowly or if its decay produces a longer path length to a given detector element, the charged products will arrive at a tracker detector late, so we can identify this track as displaced by timing information.

For LLPs that decay to at least two charged particles, we can reconstruct a "dispaced vertex" from two or more displaced tracks by reconstructing the common origin of the tracks (if an LLP decays inside a calorimeter, it may only form a displaced jet but not a displaced vertex). The situation is similar to decays including two or more photons. Once we get the displaced vertex, , its invariant mass can be calculated from all tracks originating from the vertex $\sqrt{(E_{vtx}^{tot})^2 - (p_{vtx}^{tot})}$.

5 Conclusion

In this report, we have surveyed the theoretical landscape and experimental challenges associated with Long-Lived Particles (LLPs). Our review highlights that LLPs are generic predictions in many well-motivated BSM extensions, arising from mechanisms such as kinematic suppression or weak interactions with the SM sector. We discussed how the complexity of these theories can be effectively managed using simplified models, which categorize signatures based on distinct production and decay topologies.Furthermore, this report summarized the specialized detection techniques required at the LHC. We conclude that standard searches for promptly decaying particles are insufficient for LLPs. Instead, identifying these particles requires exploiting specific detector capabilities, such as measuring anomalous energy loss (dE/dx) for charged tracks or reconstructing displaced vertices for neutral decays. As the search for new physics continues, focusing on these unconventional signatures remains a crucial strategy for probing the lifetime frontier.

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